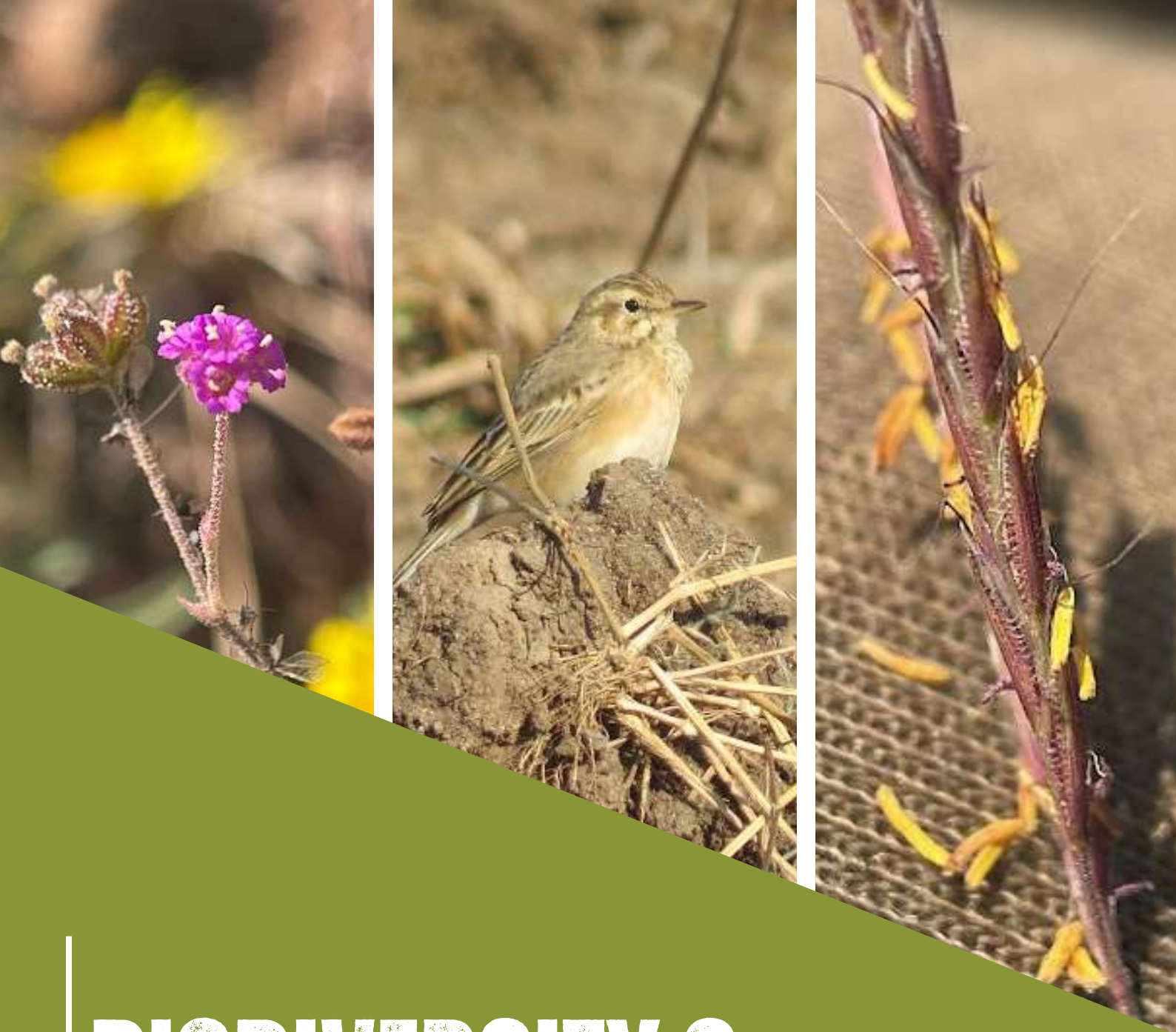




BIODIVERSITY & RESEARCH GROUP REPORT FY 2025-26



PRESENTED BY
14 TREES FOUNDATION



EXECUTIVE SUMMARY

14 Trees Foundation has been engaged in reviving the socio-ecological health of degraded lands in Maharashtra since 2013. Our goal is not merely to restore ecosystems to a historical state, but to develop healthy ecosystems capable of withstanding the pressures of rapid climate change and increasing developmental stress in the future.

A healthy ecosystem is governed by healthy ecological functions — the vital processes that keep ecosystems stable, productive, and functioning over time. These functions regulate the flow of energy, nutrients, water, and biological interactions that collectively sustain ecosystem health and resilience.

The visible outcomes of a healthy ecosystem include improvements in biodiversity, water availability, and local microclimate conditions. Since the beginning of our restoration work in 2013, we have consistently observed gradual ecological improvements across our sites. From the past year onwards, we began systematically investigating the underlying ecosystem processes driving these positive trends.

Our first focus in FY 2025-26 has been **ecosystem productivity** — the process through which ecosystems capture solar energy and convert it into chemical energy stored as biomass. A pilot study was initiated at our model restoration site in Vetale to assess productivity through carbon sequestration measurements. From FY 2026–27 onwards, the same study design will be expanded to ten additional gairans to generate long-term ecological datasets across multiple restoration sites.

In parallel, baseline ecological data were collected from four additional gairans where plantation activities will begin in FY 2026–27. These sites will be periodically monitored to evaluate the ecological impacts of restoration over time.



EXECUTIVE SUMMARY

We also began working with volunteers to identify potential reference models — tentative target ecosystems for the landscapes where we operate. This effort aims to generate insights into the socio-ecological characteristics of these ecosystems prior to degradation, helping us define clearer ecological directions and restoration goals for the future.

The expanding work of the Biodiversity & Research Group at 14 Trees Foundation is being strengthened by a growing network of volunteers who are contributing their time and energy towards positive climate action, as well as by the continued support of donors who have consistently encouraged and enabled us to expand our work.

Finally, none of this work would be possible without the support of our on-site staff, comprising members of tribal and local communities. Their contribution to field data collection, restoration activities, and logistical coordination remains central to the successful implementation of our work on the ground.



CORE TEAM MEMBERS



Pravin Bhagwat
(Founder Director)



Nirmala Samant
(Board Member)



Medha Bhagwat
(Creative Partner)



Lily Jain
(Partnership
Manager)



Anant Tayade
(Site Manager)



Sheetal Kulkarni
(Operations & Tech
Manager)



Ganesh Jagdale
(Volunteer Manager)



Abhishek Singh
(IT Manager)



Yogita Karpate
(Ecologist)

SUPPORTING TEAM



Sanket Poshire
(backoffice team)



Vinayak Naidu
(Backoffice team)



Anushka Mankumbare
(Backoffice team)



Vaishnavi Payagude
(Backoffice team)



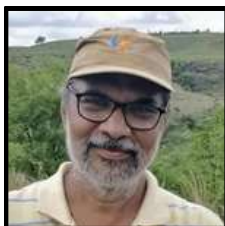
Abha Kulkarni
(intern)



Amar Khandagale
(Field Assistant)



VOLUNTEERS



Sanjeev Jagtap



Sanjeev Naik



Poorva Joshi



Aditya Ingle



Shruti Khonde



Harshil

FUNDING PARTNERS

RM Tulpule Charitable Trust

PARTNERS



TABLE OF CONTENTS

Carbon Sequestration Assessment	8
---------------------------------	---

Gairan Biodiversity Survey	14
----------------------------	----

Reference model study	18
-----------------------	----

FY 2025-26 Scorecard	26
----------------------	----

FY 2026-27 Goalpost	30
---------------------	----





The work carried out during FY 2025–26 was supported by the **R. M. Tulpule Charitable Trust**. The Trust supports research initiatives in the fields of biodiversity and clean energy across both academic and non-academic sectors. 14 Trees Foundation is honoured to have the Trust as a funding partner and looks forward to continuing this collaboration in future endeavours.

We sincerely thank **Mr. Ajay Laddha, Autus wealth Pvt. Ltd.** for generously donating a laptop and supporting our work towards creating greater impact.



CARBON SEQUESTRATION ASSESSMENT



RESEARCH FOCUS

Carbon sequestration is a key ecosystem function through which atmospheric carbon is stored in plant biomass and soils. Healthy ecosystems regulate carbon more efficiently over time, highlighting the importance of conserving natural habitats and restoring degraded lands. In this study, we compared aboveground biomass and soil organic carbon across tree plantations, grasslands, grazing areas, natural forests, and monoculture plantations.

PURPOSE OF STUDY

Identifying suitable restoration approaches is often challenging due to unclear reference ecosystems and future environmental uncertainties. Carbon sequestration can serve as an important indicator for evaluating restoration outcomes and guiding the development of resilient socio-ecological systems.

RELEVANCE FOR 14 TF

Through this study, 14 Trees Foundation developed a framework to systematically measure carbon sequestration in plant biomass and soils. Long-term monitoring, combined with other ecological indicators such as biodiversity, water regulation, and microclimate improvement, will support more informed and holistic restoration decisions.

BENEFITS BEYOND 14 TF

This project compares land-use types based on their carbon sequestration potential. The framework developed through this study can help restoration practitioners systematically measure carbon sequestration and support policymakers in incentivizing ecological restoration.



KEY INSIGHTS



Our one-year analysis at the 14 Trees Foundation project site showed that **highest carbon is sequestered in aboveground tree biomass**, followed by understory vegetation, while soil carbon remained comparatively low across all land-use types.

As this is an exploratory assessment based on one year of data, long-term monitoring is necessary to understand the influence of climate, land-use intensity, succession stage, and other ecological factors on rate of carbon sequestration. Alongside repeated carbon assessments, future studies will also document potential drivers such as biodiversity, climate, soil temperature, soil moisture, and land-use intensity.



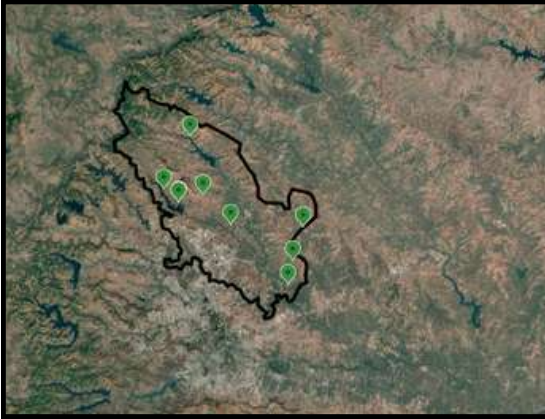
KEY INSIGHTS

Land-use	Total Area (ha)	Total CO2 eq (tonn/ha)
14 TF Plantation (< 5 years)	~ 34.43	~ 11.47
14 TF Plantation (> 5 years)	~ 5.93	~ 23.07
Grazing	~ 16.55	~ 9.89

A soil survey was conducted across the Vetale site in April 2025 in collaboration with Oikos for Ecological Services to assess Soil Organic Carbon (SOC) across different land-use types (Plantation, Wild forests, Grasslands etc.). **Preliminary results showed higher SOC in plantation areas** compared to grazing lands, with **older plantations (>5 years) consistently exhibiting greater carbon accumulation under protected conditions**. Burned areas were excluded from the analysis due to inconsistent results.



NEXT STEPS



In FY 2026-27, we plan to expand our data collection framework to **new locations** within the existing project site and also expand it to new restoration locations.



We plan to innovate our reporting methodology by developing a **real-time data monitoring and analytics dashboard**. We are also revamping our in-house app to support on-ground ecological surveys. Data collected through the app will pass through an automated processing pipeline, enabling near real-time analysis and visualization on the dashboard.





GAIRAN BIODIVERSITY SURVEY



LOCATIONS

In FY 2025–26, we completed a **baseline survey of four Gairans** (community grazing lands), where we recorded observations of flora and fauna in collaboration with **EcoConnect**. As plantation activities had not yet begun at these four sites, the data collected can largely be considered baseline observations. However, it is important to note that pit digging had already been carried out prior to the survey. Therefore, while the dataset may not represent a perfectly undisturbed baseline, it remains sufficiently robust for future impact assessment and comparative ecological analysis.

We plan to periodically monitor these sites following plantation activities to document ecological changes and restoration progress over time. The following tables presents the number of species of flora and fauna recorded during the survey.

In addition to collecting baseline data from newly acquired sites, in FY 2026-27 we will **systematically survey ten gairans for biodiversity, soil, and carbon sequestration assessment**.



FLORA

GAIRAN NAME	HERB SPECIES	GRASS SPECIES	TREE SPECIES	SHRUB SPECIES	CLIMBER SPECIES
Retwadi Bailgada Ghat, Gairan	21	11	13	4	2
Retwadi Mallinath Tekadi Gairan	24	16	7	3	1
Kanhewadi	21	6	11	2	3
Savardari	28	22	8	1	-

FAUNA

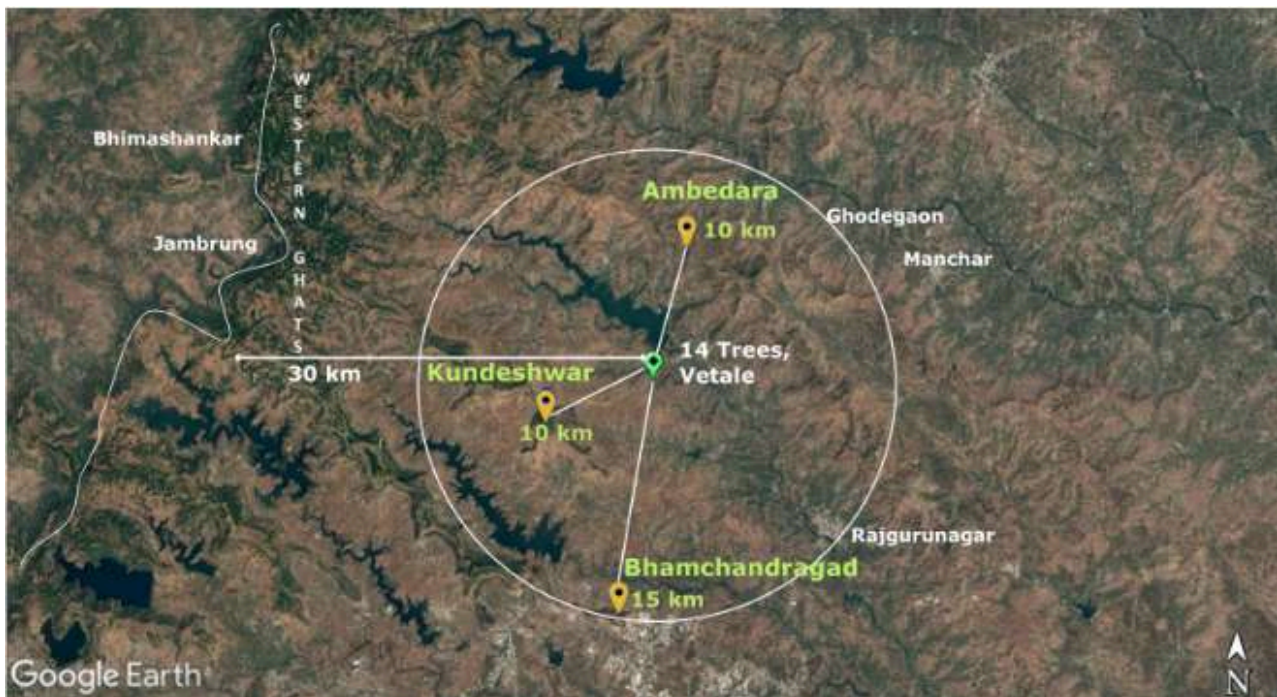
GAIRAN NAME	BIRDS	INSECTS	MAMMALS
Retwadi Bailgada Ghat, Gairan	10	14	1
Retwadi Mallinath Tekadi Gairan	7	12	1
Kanhewadi	40	11	-
Savardari	11	11	1





REFERENCE MODEL STUDY





In FY 2025–26, we initiated the mapping of reference models—target ecosystems that can guide restoration planning—for our model site in Vetale. Reference sites were identified by mapping natural and relatively undisturbed ecosystems within a 10–15 km radius of the restoration site. For Vetale, we identified three potential reference sites: **Kundeshwar** (already mapped), **Ambedara**, and **Bhamchandragad**. Kundeshwar survey was done in collaboration with **Oikos for Ecological Services**.

Since these reference ecosystems are also not entirely intact, we are complementing field observations with historical literature and remote sensing analyses to understand the historical ecological conditions of the degraded landscapes where we work. Conducting this exercise at a larger landscape scale will help us better understand past ecosystem states and guide the planning of ecologically appropriate restoration interventions.

BASELINE SURVEY OF KUNDESHWAR - FLORA

S.NO.	SCIENTIFIC NAME	REGIONAL NAME	HABBIT
1	Pongamia pinnata	करंज	T
2	Syzygium cumini	जांभूळ	T
3	Acacia torta	चिलार	Cl
4	Combretum albidum	पिळूक	Cl
5	Bauhinia racemosa	आपटा	T
6	Dalbergia latifolia	शिसम	T
7	Diospyros montana	गोयंडा	T
8	Eriolaena quinquelocularis	बोथी	T
9	Erythrina stricta	काटे पांगारा	T
10	Ficus arnotianna	पायर	T
11	Ficus racemosa	उंबर	T
12	Grewia sp.	धामण	T
13	Heterophragma quadriloculare	वारस	T
14	Lagerstroemia microcarpa	नाणा	T

*T represents Trees, Cl represents Climbers

BASELINE SURVEY OF KUNDESHWAR - FLORA

S.NO.	SCIENTIFIC NAME	REGIONAL NAME	HABBIT
15	Radermachera xylocarpa	खडशिंगी	T
16	Terminalia bellirica	बेहेडा	T
17	Terminalia chebula	हिरडा	T
18	Ziziphus rugosa	तोरण	T
19	Artemisia nilagirica	दवणा	S
20	Calotropis gigantea	मांदार	S
21	Carvia callosa	कारवी	S
22	Colebrookea oppositifolia	बामणी	S
23	Gnidia glauca	रामेठा	S
24	Lepidagathis cuspidata	काटे अडुळसा	S
25	Maytenus rothiana	भालवंड	S
26	Pavetta crassicaulis	फापट	S
27	Rhinacanthus nasutus	गजकर्णी	S

*T represents Trees, S represents Shrubs

BASLINE SURVEY OF KUNDESHWAR - FLORA

S.NO.	SCIENTIFIC NAME	REGIONAL NAME	HABBIT
28	<i>Solanum indicum</i>	चिचार्डी	S
29	<i>Spermadictyon suaveolens</i>	जीतसाया	S
30	<i>Vernonia divergens</i>	बंदर	S
31	<i>Vitex negundo</i>	निरगुडी	S
32	<i>Woodfordia fruticosa</i>	धायटी	S
33	<i>Carissa congesta</i>	करवंद	L
34	<i>Arisaema sp.</i>	सापकांदा	H
35	<i>Canscora diffusa</i>	किलवर	H
36	<i>Celosia argentea</i>	कुई	H
37	<i>Chlorophytum glaucum</i>	फुलकाडी	H
38	<i>Commelina sp.</i>	केना	H
39	<i>Conyza stricta</i>	बट दवणा	H
40	<i>Curcuma pseudomontana</i>	रान हळद	H

*T represents Trees, S represents Shrubs, H represents Herbs, G represents Grasses, L represents Lianas, Cl represents Climbers, CS represents Climbing Shrubs, and O represents Orchids.

BASELINE SURVEY OF KUNDESHWAR - FLORA

S.NO.	SCIENTIFIC NAME	REGIONAL NAME	HABBIT
41	Echinops sahyadricus	काटे चेंडू	H
42	Ensete superbum	चवेणी	H
43	Echinops sahyadricus	काटे चेंडू	H
44	Ensete superbum	चवेणी	H
45	Zingiber sp.	रान आले	H
46	Acacia torta	चिलारवेल	CS
47	Momordica dioica	करटोली	Cl
48	Paracalyx scariosus	रान घेवडा	Cl
49	Porana racemosa	भोवरी	Cl
50	Rubia cordifolia	मंजिष्ठ	Cl
51	Solena heterophylla	गोमेटी	Cl
52	Trichosanthes bracteata	कवंडळ	Cl
53	Tylophora sp.	—	Cl
54	Vigna vexillata	हळुंदा	Cl

*T represents Trees, S represents Shrubs, H represents Herbs, G represents Grasses, L represents Lianas, Cl represents Climbers, CS represents Climbing Shrubs, and O represents Orchids.

BASELINE SURVEY OF KUNDESHWAR - FAUNA

Sr. No.	Scientific Name	Common Name	Taxa
1	Accipiter badius	Shikra	Birds
2	Halcyon smyrnensis	White-throated Kingfisher	Birds
3	Moth sp. (To be identified)	Moth	Insects
4	Polyura athamas	Common Nawab	Insects
5	Hypolimnas misippus	Danaid Eggfly	Insects
6	Junonia iphita	Chocolate Pansy	Insects
7	Rhene sp.	Jumping Spider	Insects
8	Red Spider Mite	Red Spider Mite	Insects
9	Tabanus sp.	Horsefly	Insects





FY 2025-26 SCORECARD



In FY 2025–26, we committed to scaling our activities and expanding into new research areas. Below is an update on the progress made against these commitments.



Enhancing Ecological surveys further

In FY 2025–26, we systematically collected data on carbon sequestration and documented floral diversity across all seasons in permanent plots at our model site in Vetale. However, due to limited team capacity, we were unable to systematically monitor faunal diversity across the project site.



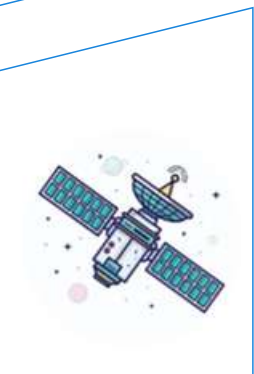
Baseline data collection for newly acquired Gairans

We acquired baseline data from four gairans where the plantation activity is planned in monsoons of FY 2026-27.



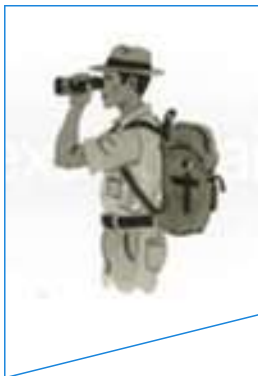
Allignment with TGBS standards

We are aligning our data collection efforts with the TGBS framework by adopting the prescribed parameters and methodologies. Going forward, we plan to expand data collection to include additional ecological parameters in alignment with TGBS standards.



Remote sensing & GIS

We used Remote Sensing and GIS techniques to assess carbon sequestration patterns across our project site using indices such as NDVI, NDMI, NDWI, and NPP. Additionally, we carried out Land Use Land Cover (LULC) mapping for our model site.



Systematic Surveys in Gairans

We had planned systematic data collection on biodiversity, soil, and land-use across gairans where plantation activities were completed or ongoing. However, due to limited team capacity, this activity could not be completed within FY 2025–26. To address this gap, surveys of these gairans have already begun in FY 2026–27.



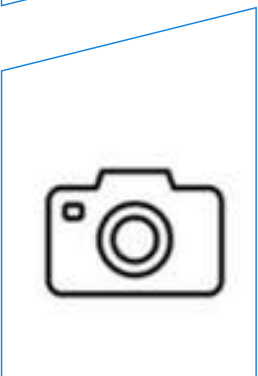
Insect diversity mapping

We initiated moth surveys and planned to expand assessments to document broader insect diversity. However, due to limited in-house expertise in insect identification, this activity could not be completed. This gap will be addressed in FY 2026–27 through a partnership with INHER.



Bioacoustics

We initiated passive acoustic monitoring by placing acoustic recorders at different locations to capture bird calls. In FY 2026–27, we plan to expand this data collection to additional locations, including gairans, and begin systematic analysis of the recordings.



Camera Trapping

In FY 2025–26, we observed signs of leopard activity along with evidence of other predators across our model site and nearby gairans. To systematically document this activity, we planned to deploy camera traps; however, due to limited resources, this activity has been postponed to the next financial year.



Socio-economic data

We planned to collect data on socio-economic parameters to understand how our interventions are influencing local socio-economic dynamics. During FY 2025–26, we began developing a framework for this assessment, which is likely to be implemented in FY 2026–27 through partnerships with social scientists.



Biochar experiments

We conducted a feasibility assessment for biochar generation and found that the biomass available within our project sites was not suitable from a biochar perspective. Since biochar production is more viable using agricultural waste, which falls beyond the scope of our current operations, we decided not to pursue this activity further.



FY 2026-27 GOALPOST



From FY 2026–27 onwards, we plan to build upon the developments initiated in FY 2025–26 and continue progressing on ongoing activities. We will continue partnering with experts and organizations to work towards the ecological and restoration goals identified for FY 2026–27. We also plan to collaborate with organizations on specific projects and engage more volunteers and students in our work.



Enhancing Ecological surveys further

In FY 2026–27, we will implement our data collection framework across additional locations within our model site, as well as in gairans, to systematically document seasonal patterns in biodiversity, soil health, and land-use change.



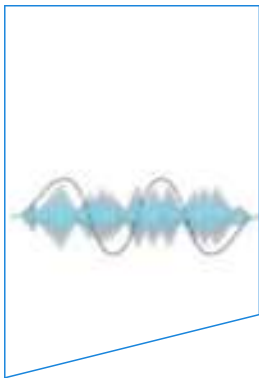
Systematic Surveys of Gairans

In FY 2026–27, we plan to conduct systematic seasonal surveys across ten gairans. Through these surveys, we will collect data on biodiversity, soil, water, and land-use patterns.



Insect diversity mapping

In FY 2026–27, we plan to map seasonal insect diversity within our model project site in Vetale. This activity will help us understand how biodiversity is responding to changes in land use.



Bioacoustics

In FY 2026–27, we will begin analyzing recordings from our acoustic recorders and deploy additional recorders across ten gairan sites to collect seasonal soundscape data. This data will be analyzed to better understand patterns of bird diversity.



Remote Sensing

In FY 2026–27, we will incorporate remote sensing and advanced analytics using machine learning models for the supervised classification of land-use and land-cover patterns, aboveground biomass, and soil characteristics.



Socio-economic mapping

We will collaborate with social scientists to better understand the socio-ecological dynamics of the tribal communities we engage with. This will provide deeper insights into how nature-rooted cultural practices can be integrated into our restoration model to support sustainable livelihoods.



Conservation of IUCN Red listed species

From FY 2026–27, our emphasis is more on conservation of red listed flora on IUCN portal. We are planning to increase the proportion of trees and shrubs from the critically endangered, threatened, and vulnerable categories in our restoration sites.

THANK YOU!

We would like to express our sincere gratitude to all collaborators, funding partners, and on-site staff for their continued support. Special thanks to all volunteers and core team members at 14 Trees Foundation, whose enthusiasm and dedication have continuously supported the growth of our research for restoration.

We are also excited to share that our volunteer network is steadily growing, and we look forward to collectively strengthening our efforts towards meaningful climate action in FY 2026-27.



Site Address

14 Trees Foundation
Vetale, Taluka Khed
Pune - 412404
Maharashtra

Pune Office Address

14 Trees Foundation
Sr. No. 149/2A, Plot No. 3, 2nd Floor,
Vishwakalyan Association of Apartments,
Near Parihar Chowk, Aundh, Pune, Haveli,
Maharashtra, India, 411007

Contacts

General inquiries: contact@14trees.org +91-9096531278
Partnerships and Corporate: partner@14trees.org +91-7249590014
Research & biodiversity group: yogita.karpate@14trees.org +91-7042299858